Ravi Teja Akula

Vijayawada, India · akularaviteja2004@gmail.com Phone private LinkedIn · Github

Education

Bennett University Greater Noida, India
Bachelor of Technology, Computer Science Engineering, Artificial Intelligence Specialization 2023 – 2027

Work Experience

Amenity Forge Hyderabad, India
ML & Backend Engineer Intern Sep 2025 – Jan 2026

- Developed machine-learning and backend modules using Python, TensorFlow, Scikit-learn, Node.js, and MongoDB, integrating machine learning capabilities into AI-driven software products
- Built RESTful APIs and data preprocessing pipelines for ML applications, enabling reliable data flow between ML models and backend services
- Performed feature engineering, model evaluation, and hyperparameter tuning, improving model accuracy by 8–12% over the baseline.
- Contributed to end-to-end product development, including model integration, API testing, debugging, deployment, and workflow automation

Academic Projects

AI Healthcare Platform Jan 2026

- Implemented TensorFlow/Keras predictive models, NLP pipelines, and Google Gemini LLM with Node.js, Express.js, React.js, and MongoDB to deliver AI-powered health assessments and personalized recommendations
- Implemented Random Forest and XGBoost models for migraine and cardiovascular risk prediction, achieving 92% and 94% accuracy, respectively, with secure JWT authentication and RESTful APIs
- Integrated LLM-based symptom analysis, personalized diet planning, and real-time health monitoring into a scalable architecture

Menstrual Migraine Predictor Feb 2025

- Developed a machine learning system to distinguish menstrual migraines from non-menstrual migraines using clinical symptom data
- Performed data preprocessing, feature selection, model training, and hyperparameter tuning using Scikit-learn, improving model accuracy from 89% to 94% and achieving an F1-score of 0.93 on the test set
- Designed and implemented a React.js and Tailwind CSS web interface enabling users to enter symptoms and receive real-time AI-powered health predictions through RESTful APIs

WhatsApp Chat Analyzer May 2026

- Built a Python-based WhatsApp Chat Analytics platform using Streamlit and Pandas to transform raw chat exports into structured datasets and actionable communication insights
- Engineered preprocessing pipelines for message extraction, timestamp analysis, user segmentation, multilingual text cleaning, and custom stopword removal to improve text-analysis quality
- Developed visual analytics with Matplotlib, WordCloud to identify user activity trends, peak engagement periods, emoji usage patterns, etc.
- Designed an interactive Streamlit dashboard to explore conversation statistics, user-level contributions, temporal activity, and text-frequency patterns

Publication

- Optimized MM-CoLLM: A Multimodal Framework for Product Recommendation | Presented at the 4th International Conference on Networks and Cryptology at Jawaharlal Nehru University, New Delhi (Under Review)

Leadership Experience

Amenity Forge Hyderabad, India
Internship Division Head Sep 2025 – Jan 2026

- Coordinated interns, assigned responsibilities, tracked progress, and facilitated communication between the interns and the organization
- Supported interns through their projects by organizing technical tasks, addressing challenges, and ensuring timely completion of deliverables

Company Event Volunteer Sep 2025 – Jan 2026

- Planned and executed company events, supporting event logistics, participant coordination, and on-ground activities
- Collaborated with team members to ensure smooth event execution and coordination among participants and organizers

Awards and Achievements

- Ranked 26th among ~700 teams in the Smart India Hackathon internal selection at Bennett University (September, 2025)
- Represented Bennett University at the national level in the Smart India Hackathon (October, 2025)

Certifications

- AWS Certified Machine Learning Associate | October 2026
- Algorithmic Toolbox – UC San Diego | March 2025
- Introduction to Machine Learning – Duke University | October 2024
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization – DeepLearning.AI | March 2025
- Databases and SQL for Data Science with Python – IBM | October 2024
- Build Basic Generative Adversarial Networks – DeepLearning.AI | March 2026

Skills and Interests

Technical Skills: Python, Java, C++, SQL, HTML, CSS, JavaScript, TensorFlow, Keras, Scikit-learn, React.js, Node.js, Express.js, MongoDB, REST APIs, NumPy, Pandas, Matplotlib, Git, GitHub, VS Code, PyCharm, Jupyter Notebook, Google Colab
Other Skills: Machine Learning, Deep Learning, Artificial Intelligence, Natural Language Processing, Generative AI, Data Science